Ex.No.:1

Installation and Exploration of the Packet Analyzer Wireshark Tool

Aim:

To install and explore different facilities available in Wireshark.

About Wireshark:

Wireshark is a powerful and widely used packet analyzer (also known as a network protocol analyzer) that allows we to capture, analyze, and inspect network traffic. It is an open-source tool available for various platforms, including Windows, macOS, and Linux.

Installation:

1. Visit the official Wireshark website: https://www.wireshark.org/
2. Go to the "Download" section and select the appropriate installer for wer operating system.
3. Follow the installation instructions provided by the installer.

Exploring Wireshark:

Once Wireshark is successfully installed, follow these steps to explore the tool:

1. **Launching Wireshark:**
   - On Windows: Look for the Wireshark shortcut on our desktop or start menu and launch the application.
   - On macOS: Open Finder, go to Applications and find Wireshark in the list of applications. Double-click to open it.
   - On Linux: Launch Wireshark from the applications menu or use the terminal command wireshark.
2. **Selecting the Network Interface:**
   Upon launching Wireshark, it will prompt we to choose a network interface to capture packets. Select the appropriate interface connected to the network we want to analyze.
3. **Capturing Packets:**
   Once the interface is selected, Wireshark will start capturing packets in real-time. We will see a list of captured packets displayed in the main window.
4. **Analyzing Packets:**
   We can click on individual packets in the list to expand and view detailed information about each packet, such as source and destination addresses, protocols used, packet size, and more.
5. **Filtering Packets:**
   Wireshark allows us to apply filters to focus on specific types of packets or traffic. We can use display filters to narrow down the packets shown based on criteria like IP addresses, protocols, port numbers, etc.
6. **Protocol Hierarchy:**

Wireshark provides a protocol hierarchy view, allowing us to see the breakdown of protocols used in the captured packets. This can help us identify which protocols are being utilized more heavily.

7. **Follow TCP Stream:**
   For TCP traffic, we can right-click on a packet and select "Follow TCP Stream" to see the entire conversation between the client and server in a more readable format.
8. **Save Captures:**
   We can save our captured packets to a file for further analysis or to share with others. To do this, go to "File" > "Save" or "Save As" and choose a file format (e.g., pcap).

The key pieces of information typically available for each captured packet in Wireshark:

- **No.:** A sequential number assigned to each packet as it is captured.
- **Time**: The time when the packet was captured, including the date and timestamp.
- **Source:** The source IP address and port number (if applicable) of the packet.
- **Destination:** The destination IP address and port number (if applicable) of the packet.
- **Protocol:** The network protocol used in the packet (e.g., TCP, UDP, ICMP, HTTP, etc.).
- **Length:** The length of the packet, including both the header and payload data.
- **Info:** A brief summary or description of the packet, which may include additional details like HTTP requests or DNS queries.

We can gain insights into various aspects of network communication by capturing and analyzing these packets including:

- Identifying the source and destination of network traffic.
- Analyzing the protocols and services being used on the network.
- Troubleshooting network issues and performance bottlenecks.
- Detecting and investigating network security threats and anomalies.
- Examining application-specific data transmitted over the network.

Some common types of packets that Wireshark can capture:

- **Ethernet Frames**: These are the basic units of data transmission in Ethernet networks. They contain source and destination MAC addresses, as well as payload data.
- **ARP Packets**: Address Resolution Protocol (ARP) packets are used to map IP addresses to MAC addresses in a local network.
- **IP Packets**: Internet Protocol (IP) packets carry data across IP-based networks. They include source and destination IP addresses, as well as payload data.

- **ICMP Packets**: Internet Control Message Protocol (ICMP) packets are used for network diagnostics, error reporting, and management functions (e.g., ping requests and replies).
- **TCP Packets**: Transmission Control Protocol (TCP) packets are connection-oriented and provide reliable data delivery with acknowledgment and retransmission capabilities.
- **UDP Packets**: User Datagram Protocol (UDP) packets are connectionless and are often used for real-time applications where reliability is less critical.
- **HTTP Packets**: Hypertext Transfer Protocol (HTTP) packets are used for web browsing, and they carry data related to HTTP requests and responses.
- **DNS Packets**: Domain Name System (DNS) packets are used for translating domain names into IP addresses.
- **DHCP Packets**: Dynamic Host Configuration Protocol (DHCP) packets are used for automatic IP address allocation within a network.
- **FTP Packets**: File Transfer Protocol (FTP) packets are used for transferring files over a network.
- **SSH Packets**: Secure Shell (SSH) packets are used for secure remote access to systems.
- **SMTP Packets**: Simple Mail Transfer Protocol (SMTP) packets are used for sending email messages.
- **POP3 Packets**: Post Office Protocol version 3 (POP3) packets are used for retrieving email messages from a server.
- **IMAP Packets**: Internet Message Access Protocol (IMAP) packets are used for managing and retrieving email from a server.
- **TLS/SSL Packets**: Transport Layer Security (TLS) and Secure Sockets Layer (SSL) packets are used for encrypted data transmission.
- **VoIP Packets**: Voice over Internet Protocol (VoIP) packets are used for transmitting voice data in real-time communication.
- **SNMP Packets**: Simple Network Management Protocol (SNMP) packets are used for network monitoring and management.

Result:

Thus, a study on various features provided by Wireshark for capturing and analyzing different types of packets has been completed successfully.